Lab: Mapping Your Invention's Innovation Ecosystem

Objective

Map the complete innovation ecosystem surrounding your invention (or a chosen invention), identifying all key components, boundaries, feedback loops, and dynamics using Active Inference systems analysis. By the end of this lab, you will have a comprehensive ecosystem map that reveals strategic opportunities and constraints.

Materials and Prerequisites

- Your invention concept from previous modules (or select a new one)
- Blank paper or digital whiteboarding tool for ecosystem mapping
- Access to basic market research resources (web search, industry reports)
- Familiarity with Markov blanket concepts from Section 1

Part 1: Identify Ecosystem Components (15 minutes)

Begin by listing every entity, institution, technology, and force that interacts with your invention. Do not filter — cast a wide net.

Step 1: Name your invention and its core function.

Write your response here

Step 2: Fill in the ecosystem component table. List at least 3 entries per category.

Category	Component	Role in Ecosystem	Relationship to Your Invention
Users/Customers			
Users/Customers			
Users/Customers			
Competitors			
Competitors			
Competitors			
Suppliers/Partners			
Suppliers/Partners			
Suppliers/Partners			
Regulators/Standards			
Regulators/Standards			
Complementary Products			
Complementary Products			
Infrastructure			
Infrastructure			

Write your response here

Step 3: Identify any components that do not fit neatly into the categories above. What does their existence tell you about the uniqueness of your ecosystem?

Write your response here

Part 2: Draw the Markov Blanket (15 minutes)

Now organize your components into an Active Inference systems framework.

Step 1: Classify each ecosystem component as belonging to one of four categories:

Component	Internal State	Sensory State	Active State	External State

Remember: - **Internal states**: Things within your invention/organization that you directly control - **Sensory states**: Channels through which you receive information from the ecosystem - **Active states**: Channels through which you influence the ecosystem - **External states**: Factors that affect your invention but that you cannot directly control

Step 2: Identify the three most critical sensory states — the information channels where losing access would most increase your uncertainty.

Write your response here

Step 3: Identify the three most impactful active states — the actions through which you most strongly influence the ecosystem.

Write your response here

Part 3: Analyze Ecosystem Dynamics (15 minutes)

Step 1: Identify the ecosystem phase. Based on the lifecycle model (emergence, growth, maturity, decline/transformation), where is your invention's ecosystem currently positioned?

Write your response here

Justify your assessment with at least three pieces of evidence:

Evidence	What It Indicates

Step 2: Identify feedback loops. List at least two positive (reinforcing) and two negative (balancing) feedback loops in your ecosystem.

Positive feedback loops (things that accelerate growth or decline): 1.

Write your response here

1.

Write your response here

Negative feedback loops (things that stabilize the system): 1.

Write your response here

1.

Write your response here

Step 3: Platform analysis. Does your ecosystem contain a dominant platform? If so, answer the following:

Question	Your Analysis
What is the platform?	
Who controls it?	
What shared generative model does it provide?	
Is your invention building on the platform, competing with it, or building a new one?	
What are the risks of platform dependence?	

Write your response here

Part 4: Regulatory and Standards Landscape (10 minutes)

Step 1: List every regulation, standard, or legal framework that applies to your invention.

Regulation/ Standard	Governing Body	How It Affects Your Invention	Constraint or Opportunity?

Step 2: Is the regulatory environment for your invention stable, evolving, or absent?

Write your response here

Step 3: If you could influence one regulatory or standards outcome, what would it be and why?

Write your response here

Part 5: Strategic Synthesis (10 minutes)

Step 1: Based on your ecosystem analysis, identify the top three opportunities for your invention.

Opportunity	Why It Exists (Ecosystem Logic)	What Action It Suggests

Step 2: Identify the top three ecosystem risks.

Risk	Why It Exists (Ecosystem Logic)	How You Could Mitigate It

Step 3: Write a one-paragraph "ecosystem strategy statement" that summarizes how your invention will position itself within its ecosystem, which components it will prioritize, and how it will navigate the current ecosystem phase.

Write your response here

Discussion and Debrief

1. How did mapping the full ecosystem change your understanding of your invention's challenges and opportunities?
2. Were there ecosystem components you had not previously considered? What does this blind spot suggest about your current generative model?
3. Compare your ecosystem map with a peer's. What structural similarities and differences do you notice? What might explain them?
4. How might your ecosystem map look different in 5 years? What phase transitions do you anticipate?
5. If your invention is in an emerging ecosystem, what actions could you take to shape the ecosystem's development in your favor?